

Kyung Hyun Lee

Clinical AI Researcher

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Summary

PhD student in Digital Health at SAIHST, Sungkyunkwan University. AI researcher working on clinical time-series deep learning, medical AI regulatory pathways, and respiratory digital health. Co-founder & CEO of BreathYou.

Education

PhD Student	Sungkyunkwan University (SAIHST) , Digital Health • Advisor: Prof. Byung-Jae Lee • Research on Transformer-based trajectory models of respiratory health	Seoul, South Korea 2026 – present
Master of Science	Sungkyunkwan University (SAIHST) , Digital Health • Advisor: Prof. Mira Kang • M Lab. — Statistical analysis and predictive model development using medical big data	Seoul, South Korea 2019 – 2021
Bachelor of Science	Incheon National University , Electronic Engineering & Computational Neuroscience • Advisor: Prof. Sangwon Byun • Bioelectronics Lab. — Biosignal (ECG, EMG) data collection, signal analysis, and real-time prediction model development	Incheon, South Korea 2014 – 2019

Experience

BreathYou , Co-founder & CEO Digital health startup focused on allergy and respiratory AI, developing DeepBreath CDSS and MediPipe. • Selected for 2025 Pre-startup Package • DeepBreath CDSS: Clinical decision support for predicting pulmonary function decline trajectories • MediPipe: OCR/NLP pipeline for medical record processing	South Korea 2025 – present 1 year
AITRICS , Researcher (2024 – present) · Industrial Technical Personnel — Alternative Military Service (2022 – 2024) Medical data (RWD) cleaning, analysis pipeline construction, predictive model development, and external validation. Contributed to the cardiac arrest early-warning model that obtained MFDS (Korean FDA) approval as an AI medical device, and currently leading product development of a self-developed AKI prediction model toward MFDS approval. • Research design for clinical data analysis projects • Data extraction and cleaning from multi-center hospitals • Automated data analysis pipelines using Airflow • Deep learning-based clinical time-series prediction models • Clinical simulation-based model evaluation and survival analysis • Cardiac arrest prediction model — supported MFDS approval as an AI medical device • AKI prediction model — developed the model and currently leading productization toward MFDS approval	Seoul, South Korea 2022 – present 4 years
Exosystems , Biosignal AI Research Engineer Deep learning-based EMG modeling for Muscle Quality monitoring. • RNN, CRNN-based Muscle Quality time-series prediction model development • Clinical validation-oriented AI model evaluation	South Korea May 2022 – Aug 2022 4 months

Incheon National University, Research Fellow

- EMG-based real-time hand gesture recognition algorithm R&D
- Deep learning-based ECG modeling for autonomic nervous system monitoring

Incheon, South Korea

Sept 2021 – May 2022

9 months

Samsung Medical Center & SAIHST, Research Fellow

EHR data extraction and analysis from CDW; development of ML-based CDSS.

- Data cleaning and health checkup database construction
- Text mining of unstructured medical records
- Statistical analysis and predictive/diagnostic model development

Seoul, South Korea

Dec 2018 – Aug 2021

2 years 9 months

Teaching

University of Seoul, Department of Sport Science: Lecturer · Exercise and Digital Healthcare, fourth-year undergraduate elective (Fall 2026)

Awards

Human Understanding AI Paper Competition — Top 7 & ETRI Director's Award

2022

ETRI, Ministry of Science and ICT

- Multimodal Attention-based Korean emotion recognition model combining RoBERTa (Text) and Wav2vec 2.0 (Audio)

OCR AI Training Data Hackathon — Excellence Award

2021

Ministry of Science and ICT, NIA

- AI-Assisted OCR Annotation System for pharmaceutical products (CRAFT-based)

Medical Information Analysis Expert Advanced Course — Grand Prize

2021

Ministry of Health and Welfare, Asan Medical Center

- Real-time BIS prediction model using intraoperative EEG data (25-second lookahead)

Medical AI Expert Training Course — Excellence Award

2020

Korea Human Resource Development Institute for Health & Welfare

- 30-day mortality and ICU admission prediction model using preoperative and intra-operative vital signs

STAR Program Paper Presentation — Grand Prize

2018

DSP-IN-U Smart Device Talent Development Program

- EMG-based real-time hand gesture prediction model development

Academic Service

Computers in Biology and Medicine: Peer reviewer · 4 manuscripts (2026)

Information Processing & Management: Peer reviewer · 1 manuscript (2026)

Certificates

Medical Information Analysis Expert Advanced Course

Aug 2021

Korea Human Resource Development Institute for Health & Welfare, Asan Medical Center

Google Developer Machine Learning Boot Camp

Jan 2021

Google Developers

Medical AI Expert Training Course

Dec 2020

Korea Human Resource Development Institute for Health & Welfare

Skills

Programming & ML: Python, PyTorch, TensorFlow, Scikit-learn

Data & Database: SQL (PostgreSQL), Airflow, Data Pipeline

Statistics & Analysis: R, Survival Analysis, Clinical Statistics

Web Development: FastAPI, Flask, JavaScript, HTML/CSS, AWS

Languages

Korean

Native speaker

English

Professional working proficiency

References

Prof. Sangwon Byun

Incheon National University, Department of Electronic Engineering; Bachelor's advisor
([Bioelectronics Lab.](#) · [Scholar](#))

Prof. Mira Kang

Samsung Medical Center; Master's advisor at SAIHST, Sungkyunkwan University

Prof. Jin-Young Lee

Samsung Medical Center, Health Promotion Center

Prof. Byung-Jae Lee

Samsung Medical Center, Department of Allergy and Clinical Immunology; Doctoral advisor at SAIHST, Sungkyunkwan University

Prof. Sae Young Jae

University of Seoul, Department of Sport Science; research collaborator